escape, but because it's flexible, its walls can move outward. As the children relentlessly slam into each of the box's walls—hundreds at a time, with hundreds more immediately to follow—the box steadily expands. Now, you might expect that because the walls are impermeable, the total energy embodied by the swarming children will stay fully within the expanding box. After all, where else could their energy go? Well, although a reasonable proposition, it's not quite right. There *is* some place for it to go. The children expend energy every time they slam into a wall, and much of this energy is transferred to the *wall's motion.* The very expansion of the box absorbs, and hence depletes, the children's energy.

Even though space doesn't have walls, a similar kind of energy transfer takes place as the universe expands. Just as the fast-moving children work against the inward force exerted by the box's walls as it expands, the fast-moving particles in our universe work against an inward force as space expands: They work against the inward force of gravity. And just as the total energy embodied by the children drops because it's continuously transferred to the energy of the walls as the box expands, the total energy carried by ordinary particles of matter and radiation drops becasue it is continually *transferred to gravity* as the universe expands. In short, by drawing an analogy between the inward force exerted by the box's walls and the inward force exerted by gravity (an analogy that can be established mathematically), we conclude that gravity depletes the energy in fast-moving particles of matter and radiation as space swells. The loss of energy from fast-moving particles from cosmic expansion has been confirmed by observations of the microwave background radiation.[26]

Let's now modify our analogy a bit to gain insight into how an inflaton field impacts our description of energy exchange as space expands. Imagine that a few pranksters among the children hook up a number of enormous rubber bands between each of the opposite, outward-moving walls of the box. The rubber bands exert an inward, negative pressure on the box walls, which has exactly the opposite effect of the children's outward, positive pressure; rather than transferring energy to the expansion of the box, the rubber bands' negative pressure "saps" energy from the expansion. As the box expands, the rubber bands get increasingly taut, which means *they* embody increasing amounts of energy.